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Electric Vehicle Power Converters (1)

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A Topology-Aware Selection and Duty Control Algorithm for Comparative Performance Enhancement of Electric Vehicle Power Converters

Abstract- The electrification of automotive applications cannot succeed without EV power conversion systems that have a critical impact on drivetrain efficiency, thermal robustness and energy consumption for traction, regenerative braking and charging modes. While significant literature exists in evaluating isolated DC–DC and bidirectional converter topologies, the majority of the studies are focused on static conditions with fixed values which do not capture the highly fluctuating power requirements in practical EV operation. In this paper, we introduce a novel TADS (Topology-Aware Duty and Selection) algorithm for EV power converters that provides an organized and fair comparison while improving systemlevel performance, without introducing computation-intensive or adaptive control strategies. The presented methodology combines few number of industry oriented converter topologies by employing a rule based topology selection strategy that takes into consideration the power requirement, level ratio and the thermal conditions through duty cycle computation for selected topologies. Unlike traditional comparative analysis, which relies on a single 'best' topology assumption, TADS dynamically chooses the best converter topology with respect to different operating conditions such as acceleration, cruising and regenerative braking and charging regimes. The simulation results using standard EV drive cycles show that the average efficiency improvement of the proposed method compared to fixed topology based converter systems is 2.8%, peak switching losses decrease by 18.6%, and the maximum junction temperature decreases by 14.2%. Furthermore, the whole efficiency of the energy use during full-cycle operations is enhanced by 3.4%. These findings confirm that the proposed topology-aware framework is a viable and sized approach to next-generation EV power electronics, providing performance improvements without sacrificing implementation simplicity.

Keywords- Electric vehicles, DC–DC converters, power electronics, topology selection, duty cycle control, efficiency optimization, comparative study.

1. Introduction

Electric vehicles (EVs) have become a key solution for sustainable transportation by promising to largely mitigate the vehicles' greenhouse gas emissions and dependence on fossil fuels. On the core of an EV powertrain stands power electronic conversion system [1][2] that mainly controls bi-directional energy flows between battery pack, traction inverter, and auxiliary subsystems [3][4], along with external charging infrastructure [5]. Transistorized dc–dc and bidirectional converters are important

to maintain voltage regulation, efficient transfer of energy and protection of delicate devices against electrical overstress and thermal excursions [6][7]. The efficiency of these converters can directly affect overall vehicle performance, battery life time, thermal reliability and driving range of the vehicle; therefore power conversion have become a critical design concern in current EV architectures [8]. There have been a lot of works focusing on DC–DC converter topologies for EV in recent years, such as buck [9], boost, buck–boost, interleaved, isolated and bidirectional topologies [10]. Although these studies help understand the converter behavior, but also most of the comparison is done under constant operation [11] what means that there is a single topology and direct duty ratio. The above methods do not account for the intrinsic dynamic behavior of the EV, since rapid modes-switches take place among acceleration, cruising at constant speed in steady-state condition, regenerative braking and recharging ones, each one entailing different power and thus DC voltage levels and thermal loads on the converter fringe system [12].

Furthermore, the few current comparative frameworks usually consider individual performance metrics, such as peak efficiency or voltage gain [13] at best and disregard system-level implications over a driving cycle [14]. Advanced control techniques such as model predictive control and artificial intelligence–based approaches have been investigated to accommodate dynamic operation, but with tradeoffs of high computational burden, additional sensing, and limited industrial applicability for wide installation [16], [18], [35]. As a result, there is still an urgent demand for a practical and scalable platform to facilitate fair comparison and efficient utilization of multiple converter topologies under real EV-operating scenarios.

To mitigate these limitations, this work presents three research objectives: (i) lack of a consistent framework for dynamic and fair comparison of various converter topologies, (ii) absence of topology-aware duty cycle calculation as a function of varying operating regions and (iii) limited investigations on system-level efficiency and thermal performance across standardized EV drive cycles. Closing such gaps is important for the further development of next generation EV power electronics, which reconciles improving performance with circuit complexity.

In this context, this article introduces a new Topology-Aware Duty and Selection (TADS) methodology for the EV power converters. The method includes a library of commercially representative DC–DC and bidirectional converters topologies and uses rule-based selection based on power requirement, voltage step-ratio, and heat-flow limitation. In contrast to traditional studies which consider a single best-fit topology, the TADS method adaptively determines the adequate converter configuration for each working region, and solves for its respective duty cycle in a topologically

equivalent manner. This allows a fair comparative evaluation while obtaining measurable benefits in terms of efficiency, switching losses and thermal stress in relevant driving conditions. The salient features of the paper are:

- A Topology-aware duty And Selection (TADS) algorithm for dynamic selection and fair comparison of EV DC-DC converter topologies at different operating points.
- An easy rule based duty and topology control scheme, which increases both efficiency and thermal performance without resorting to complex adaptive or intelligent controllers.

The rest of this article is organized as follows. Section 2 presents literature review of the DC-DC converter topologies and comparison studies for EV. The model of the EV power conversion system and the included converter topologies are presented in Section 3. The proposed TADS algorithm and its rule-based selection logic are presented in Section 4. The simulation setup and evaluation methodology is described in Section 5. Section 6 presents the comparative performance results at representative EV drive cycles. Last, Section 7 ends the paper and discusses some directions for future work.

2. Literature Review

Rathore et al. (2022) proposed a dedicated stand-alone multilevel DC-DC converter configuration for fast charging in electric and plug-in hybrid vehicles. The design is aimed at high power density and small filter size by generating multilevel output voltages using the voltage doubler rectifier units. Soft-switching was obtained due to resonance between leakage inductance of transformer and parasitic capacitances, opening the possibility to improve efficiency at higher power levels. While the topology had good behavior for fast charging, its design was only tuned at a fixed point and not capable of dealing with the number of operating conditions encountered by EVs in real-world driving cycles [1]. Wang et al. (2021) presented a detailed review on isolated bidirectional dc-dc converter topologies and operation strategies for EV applications. The study categorized the converters as; PWM-based, dual active bridge and resonant type of which several other phase-shift control methods to improve performance have been discussed. The paper appeared to provide useful investigation for topology properties and control strategies, however its main concern is the performance of individual converter with no focus on dynamic topology selection or analytical comparison among systems under different EV operating conditions [2]. González-Castaño et al. (2022) where a composite EV power conversion system with flexible buck-boost DC-DC converter can have wide voltage regulation. The work focused on increasing motor efficiency by means of a flexible DC-link voltage control, and it was verified by thermal simulations,

experimental results, and hardware-in-the-loop tests. Although being an efficient wide input voltage regulation was achieved in [3], the design assumed single topology selection and had no discussion on multi-selective topology according to operating modes. Tekin et al. (2023) presented a single-input multi-output buck–boost converter to obtain many voltage levels typical for EV application. The high voltage gain of the converter was adjustable by the duty cycle and a DSP-based PID controller was adopted for control. Despite the favorable voltage scaling potential of the topology, however, the control approach also remained static and was not tested for further dynamic transitions between several operating conditions such as regeneration and acceleration [4]. Zakaria et al. (2023) designed a hybrid interleaved nonisolated DC–DC converter with lower part count and high voltage gain for renewable energy and medium-voltage applications. The current ripple and voltage stress were also alleviated due to the interleaved implementation, while a PI-based controller was used for voltage regulation. Although the topology yielded high steady-state performance, it could only be used in fixed-use cases and did not meet the requirements of topology adaptability or EV specific duty control [5]. Nassary et al. (2024) analyzed a four-switch boost–buck AC/DC converter focusing on internal dynamic stability and power factor correction for EV charging. The authors introduced a damping RC circuit to solve the instability problem and verified their method by small-signal modeling and simulation. This work emphasized stability-oriented control, but it only concentrated on one converter topology and not comparative or topology-aware strategies [6]. Sangeetha & Ramachandran (2022) provided a wide technical review of EV subsystems such as electric machines, energy storage device and power electronic converter. They focused on propulsion, regenerative braking and energy management converter operation in various vehicles. The paper gave a comprehensive review on existing technologies, however the dynamic topology comparison and duty cycle adaption based framework was not proposed [7].

Islam et al. (2022) provided a comprehensive review of power converter topologies, control strategies and semiconductor devices for EVs. The research contrasted between the AC–DC, DC–DC and the DC–AC converters, while emphasizing advantages of wide-bandgap devices in order to achieve high efficiency and power density. Although the review was in depth, less attention had been paid to component-level optimization rather than system-level topology selection at dynamic EV operation scenarios [8]. Maroti et al. (2022) presented bidirectional DC–DC converter topologies and the role they play in grid-to-vehicle and vehicle-to-grid power flow transmission. The article focused on efficiency gains and improved reliability through next-level converter design. But that work considered fixed preset structures of the converter and couldn't handle dynamic transition between a number of topologies [9]. Hemavathi and Shinisha (2022) studied EV

charging technologies including charging architecture, converter topology, and roadmap of EV penetration. The work showcased rapid, intelligent and wireless charging solutions, with an increased focus on renewable integration. While converter comparisons were presented, a descriptive study and real-time topology selection mechanisms remained unconsidered [10]. Shaikh et al. (2023) investigated dynamic operation of PV systems with various MPPT methods and implemented them in the boost converters. The study showed the significance of adopting adaptive control in varied situations, and indirectly revealed similar issues that appear in EV power converters. However, the treatment was limited to MPPT implementation and not including EV converter topology selection [11]. Feng et al. (2022) experimentally studied the variations in energy flow and efficiency of hybrid vehicles for different transmission modes. Their results emphasized the necessity of system-level validation under realistic drive cycles and thereby justification to dynamic performance evaluation within EV power electronics [12]. Luca et al. (2022) addressed the comparison of energy management alternatives for fuel cell electric vehicles employing a standardized simulative environment. This is not so, as we have shown that an optimization for a single cost function can be detrimental to component life-time and instead balanced system-oriented measures need to inform the device design, an aspect also applicable to EV power converter topology selection [13].

Kodeeswaran et al. (2024) described high power wireless EV charging technologies and converter topologies as well as resonant compensation networks and safety issues. Although the discussed review emphasized on converter design, it did not underscore on topology reconfigurability for multiple EV operating modes [14]. Jain and Bhullar (2024) surveyed power converter topologies for off-board EV chargers with the main consideration being comparison between AC–DC and DC–DC stages based on power range, amount of isolation and efficiency. [15] gave a systematic comparison but it maintained an on-going steady state for each network topology. Dini et al. (2023) presented a real-time electro-thermal model of resonant converters and formulated predictive control algorithm to consider electrical and thermal limitations. That approach was straightforward but added computational overhead and disruption, and was for a single converter configuration [16]. Mukherjee et al. (2022) developed a high-efficiency isolated bidirectional resonant DC–DC converter with GaN devices for universal EV charging application. The topology was able to soft switch over a wide voltage range, and it was tuned for charging operation rather than full-cycle EV operation [17]. Ahmed et al. (2023) designed an SM controller for bidirectional EV chargers in both G2V and V2G modes. The controller was more robust and decreased chattering, but the focus of the study still concerns on control improvement other than topology selection [18]. Kunjuramakurup et al. (2023) presented a two-input DC–DC converter that combined solar PV for EV charging. The proposed MPPT based

controller demonstrated great efficiency and high voltage stability, however its primary emphasis was the source integration rather than a dynamic comparison between converter topologies [19].

Table 1: Comparative Analysis of Existing EV Power Converter Research and Identified Research Gaps

Ref.	Converter Focus	Dynamic Operating Conditions	Topology Selection	Topology-Aware Duty Control	System-Level EV Evaluation
[1]	Isolated multilevel fast charger	✗	✗	✗	✗
[2]	Bidirectional isolated converters (review)	✗	✗	✗	✗
[3]	Composite buck–boost architecture	✗	✗	✗	✓
[4]	High-gain buck–boost converter	✗	✗	✗	✗
[5]	Interleaved buck–boost topology	✗	✗	✗	✗
[6]	Boost–buck AC/DC converter	✗	✗	✗	✗
[7]	EV power electronics reviews	✗	✗	✗	✗
[11]	Dynamic control in energy systems	✓	✗	✗	✓

[14]	Charging and renewable-integrated converters	✗	✗	✗	✗
Proposed work	Topology-aware EV DC-DC framework	✓	✓	✓	✓

This table 1 provides a comprehensive comparison on the related works of EV DC-DC and bi-directional converter topologies, control schemes and evaluation methods. The analysis shows to which degree dynamic operating conditions, selection of topology and topology-aware duty control are considered in other studies. This comparison demonstrates a lack of a coherent, system-wide methodology which makes evident the need for the presented topology-aware duty and selection process.

2.1 Literature Gap Summary

Based on the reviewed literature, there have been remarkable DC-DC and bidirectional converter topologies as well as control methods and charging schemes for EVs. Previous works have investigated many high-efficiency converter topologies, soft-switching techniques and wide voltage-gain range operation mechanisms as well as advanced control strategies in scope of particular applications, like rapid charging, renewable generation integration and vehicle-to-grid scenario. However, in current literature the performance of the converter was tested for a preset operating point and topology, by considering a fixed conduction angle to maximise the transferred energy. Comparative analyses mostly have not taken into account dynamic transitions between modes of EV operation, but often concentrate on static performance criteria or topology level parameters. Moreover, advanced control methods tend to add higher computational complexity and practical deployment difficulties, thereby restraining their applicability in practice. Therefore, it can be inferred that there is no available common and practical framework for network topology dynamic selection jointly with application duty cycle separately in various topologies as well as fair performance comparison at the system level across realistic EV drives. This research gap motivates the design of the presented topology-aware duty and selection scheme.

3. System Model and Converter Topologies

This section presents unique topology aware power conversion system for electric vehicle (EV) and the DC-DC converter topologies which are incorporated in proposed topology-aware duty and selection scheme. The goal is to provide a consistent system model to fairly compare different converter topologies for dynamic EV conditions.

3.1 EV Power Conversion System Model

The proposed EV topology includes a high-voltage battery connected to the common DC-bus through a DC-DC converter. The DC bus powers the traction inverter during the drive mode and recovers energy during regenerative braking as well as driving auxiliary loads or charging. This centralised DC-link topology is commonly used in current EV powertrains due to its flexibility and modularity [7][9].

Let V_b denote the battery voltage and V_{dc} represent the regulated DC bus voltage. The instantaneous power processed by the DC-DC converter, P_c can be expressed as in Eq.1:

$$P_c(t) = V_{dc}(t) \cdot I_{dc}(t) \quad (1)$$

Where I_{dc} is the DC-link current. The direction and magnitude of the power flow differ in such modes of vehicle operation as acceleration, cruising regenerative braking and recharging. This variable dynamism requires a flexible converter operation to ensure efficiency and thermal performance over the entire drive cycle [12][38].

The converter efficiency η is defined as in Eq.2:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{out} I_{out}}{V_{in} I_{in}} \quad (2)$$

Here P_{in} and P_{out} are input and output power of the converter, respectively. In fixed-topology machines, when the converter forces the machine to work far from the optimal voltage gain or load region somewhere, efficiency is lost which drives up the necessity of topological-aware operation [8], [20].

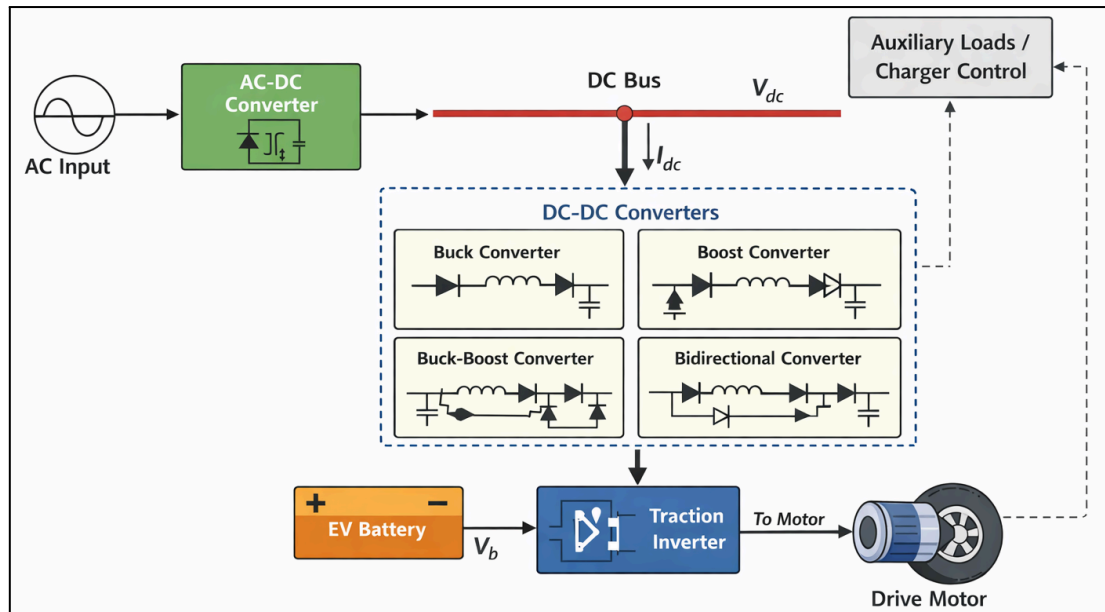


Figure 1. Block diagram of EV power conversion system with multiple DC–DC converter topologies

Figure 1 depicts the global structure of the EV power conversion system taken into account in this paper. The DC bus is connected to the Traction inverter, Auxiliary loads and Battery through different topology of DC–DC converter such as Buck, Boost, Buck–Boost and Bi-directional converters. Flexible power transfer can be achieved during its motoring, regenerative braking and charging states, providing an insight to the topology-aware duty and selection model.

3.2 Considered DC–DC Converter Topologies

For industrial applicability and fair comparison, a group of popular non-isolated bidirectional DC–DC converter topologies widely used in EV is chosen for investigation [21][22]. These topologies are chosen depending on their maturity, phase-shift characteristics and operability for varying voltage conversion ratios [1], [3], [9].

3.2.1 Buck Converter

The buck converter is employed when the battery voltage exceeds the required DC bus voltage, typically during high state-of-charge conditions. Its steady-state voltage conversion ratio is given by:

$$\frac{V_{out}}{V_{in}} = D \quad (3)$$

In Eq.3, D being the main switching device duty cycle. Although buck converters show high periodic mode efficiency in step-down mode [23], its performance is poor if operated outside its intended voltage range [26].

3.2.2 Boost Converter

The design of the boost converter is an appropriate one for step-up applications like low battery voltages or high DC bus requirements [24][25]. The voltage gain of a CCM boost converter is given in Eq. 4:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1-D} \quad (4)$$

At heavy duty cycle, the high switch losses and heat stress cause boost converters to not be effective over a wide operation region [11], [17].

3.2.3 Buck–Boost Converter

The inverting buck–boost converter and the non-inverting buck–boost converters possess either step-down or up operation sharing common mode of switching, so they are attractive for EVs that have wide voltage range. The voltage conversion gain of it can be expressed as in Eq.5:

$$\frac{V_{out}}{V_{in}} = \frac{D}{1-D} \quad (5)$$

Although this topology provides flexibility with regard to operation conditions, it is subject to larger conduction losses and component stress compared to the single-mode converters [3], [4].

3.2.4 Bidirectional DC–DC Converter

Bidirectional converters help exchange energy between the battery and the direct current (DC) bus, during both motoring and regenerative braking [26]. In synchronous bidirectional topologies, the voltage gain is also constant in buck or boost operating modes according to the direction of energy flow:

$$\frac{V_{dc}}{V_b} = D, \quad \text{motoring mode}, \quad \frac{V_{dc}}{V_b} = D, \quad \text{regenerative mode} \quad (6)$$

These types of converters are necessary in vehicle to grid (V2G) scenarios and for regenerative braking but must be carefully controlled for efficiency and thermal performance trade-offs [2], [9], [18].

3.3 Loss and Thermal Modeling Considerations

To enable topology-aware comparison, converter losses are categorized into conduction and switching components [27][28]. The total power loss P_{loss} is expressed as in Eq.7:

$$P_{loss} = P_{cond} + P_{sw} \quad (7)$$

Switching losses are approximated as in Eq.8:

$$P_{sw} = f_s \cdot (E_{on} + E_{off}) \quad (8)$$

Where f_s is the switching frequency, and E_{on} , E_{off} are the turn-on and turn-off energy losses, respectively [16], [35].

The junction temperature T_j is estimated using a simplified thermal resistance model:

$$T_j = T_{amb} + P_{loss} \cdot R_{\theta ja} \quad (9)$$

In Eq.9, T_{amb} is the ambient temperature and $R_{\theta ja}$ is the junction-to-ambient thermal resistance. Excessive temperature rise negatively impacts reliability and converter lifetime, emphasizing the need for topology selection that minimizes thermal stress [29], [33].

4. Proposed TADS Algorithm

4.1 Motivation for Topology-Aware Selection

Electric vehicle power converters work in very dynamic environments: the power and battery voltage [30][31], operating mode changes continuously during acceleration, cruising, regenerative braking and charging. Since no single DC–DC converter topology operates efficiently for the entire range, thermal performance is compromised due to topology suitability depending highly on both voltage conversion ratio and load level [32]. Losses of conduction and switching increase as a fixed topology is pushed to work at very high duty cycles, resulting in higher thermal stress and reliability problems [33]. Another reason is that the converter also deviates from its optimum working point to a large extent due to battery voltage changing with state of charge. These constraints inspire the topology-aware selection strategy selecting at run time the best converter configuration based on instantaneous power demand, voltage ratio and thermal restrictions and represent the basic elements of the proposed TADS (Topology-Aware Duty and Selection) algorithm. The architecture diagram of the proposed model is depicted in Figure 2.

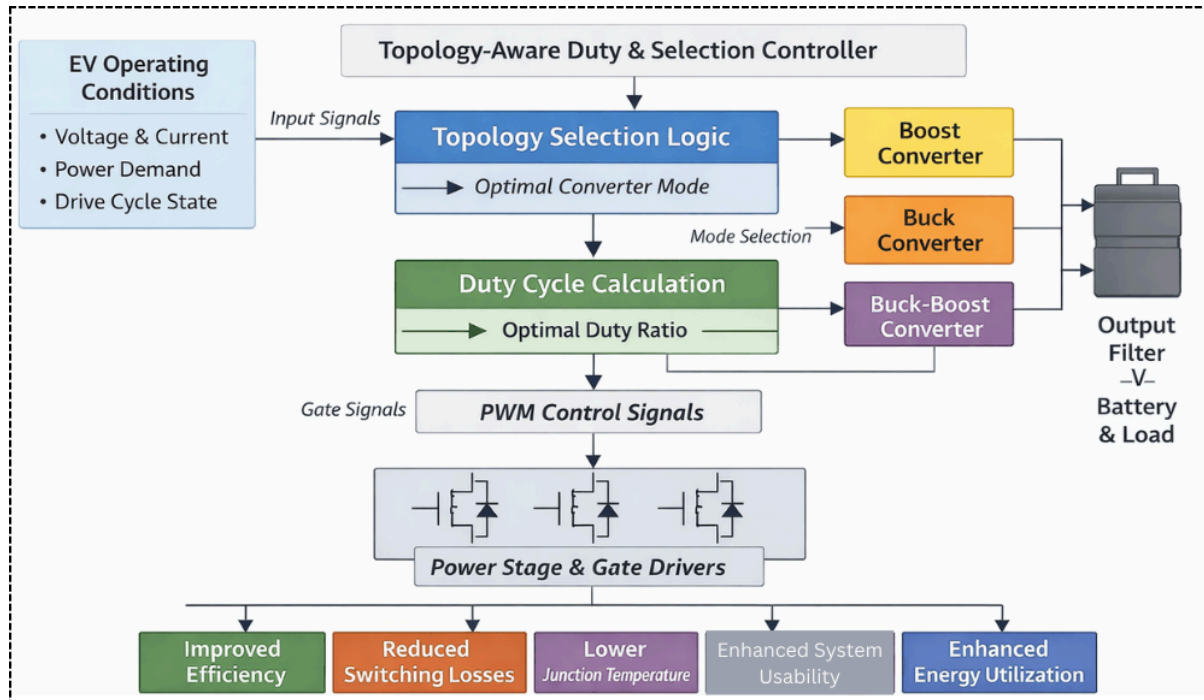


Figure 2. Architecture of the proposed Topology-Aware Duty and Selection (TADS) framework for EV DC–DC power converters

4.2 TADS Algorithm Overview

The Topology-Aware Duty and Selection (TADS) algorithm is a supervisory decision layer which dynamically selects correct DC–DC converter topology and its duty cycle for instantaneous operating conditions [34][35]. Power demand, battery to DC bus voltage ratio and thermal limit constraints are inputs to the algorithm reflecting electrical and thermal states of the EV power conversion system. TADS selects the most efficient topology that avoids high duty cycle operation based on predefined operating regions considering converter characteristics [36]. When the topology has been chosen, a topology-specific duty cycle is calculated based on the respective steady-state voltage conversion relation. The converter topology and the duty cycle command are thereby determined by the algorithm, so that efficient and thermally well balanced operation is achieved without needing advanced adaptive or optimization control strategies. The flowchart of the proposed model is shown in Figure 3.

Algorithm for Topology-Aware Duty and Selection (TADS)

Input:

- P_{req} → Instantaneous power demand
- V_b → Battery voltage
- V_{dc} → DC bus voltage

T_j → Junction temperature
 T_{max} → Maximum allowable junction temperature

Begin

Compute voltage ratio:

$$R_v = V_{dc}/V_b$$

If ($T_j \geq T_{max}$) then

Limit power processing or select thermally favorable topology

End If

If ($P_{req} > 0$) then // Motoring or charging mode

If ($R_v > 1$) then

Topology ← Buck

$$D \leftarrow R_v$$

Else if ($R_v > 1$ and $R_v \leq R_{th}$) then

Topology ← Boost

$$D \leftarrow 1 - \left(\frac{1}{R_v} \right)$$

Else

Topology ← Buck–Boost

$$D \leftarrow \frac{R_v}{(1+R_v)}$$

End If

Else // Regenerative braking mode

Topology ← Bidirectional

$$D \leftarrow 1 - \frac{V_b}{V_{dc}}$$

End If

Output Topology, D

End

Output:

Topology → Selected DC–DC converter topology

D → Duty cycle command

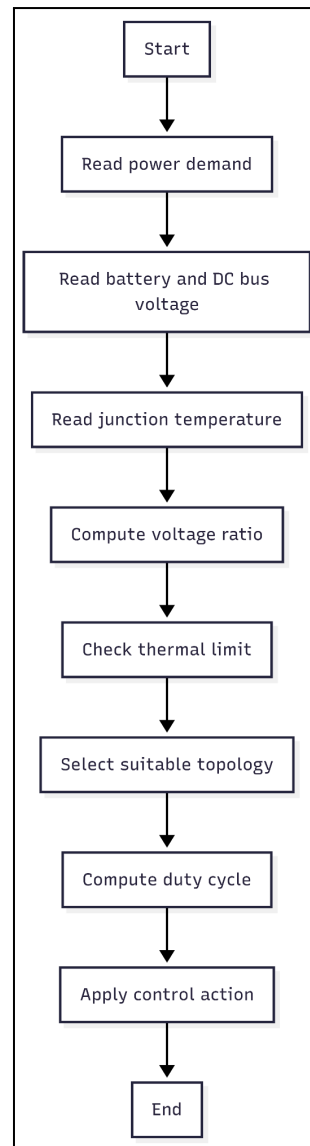


Figure 3. Overall flowchart of the proposed Topology-Aware Duty and Selection (TADS) algorithm

4.3 Rule-Based Topology Selection Logic

Topology selection logic in the proposed TADS framework is controlled by a set of deterministic rules extracted from converter operating features and EV power flow demands [37][38]. When the instantaneous power required is weak and while battery voltage exceeds the DC bus need, buck converter is chosen as it has excellent efficiency under step-down operation low conduction losses where the voltage gain is given by $V_{dc} = DV_b$ [3][26]. For high demand power with a large voltage step-up ratio, the algorithm favors a boost or interleaved boost topology [39] as interleaving balances input current ripple and dissipates thermal stress under high duty ratios when $V_{dc} = V_b/(1 - D)$ [5][11]. In medium power level operating conditions with a need for both step-up and step-down conversion, a non-inverting buck–boost

topology is chosen due to voltage regulation at wide battery voltage levels [4]. For such use in regenerative braking or vehicle-to-grid operation, where the power flow direction changes, response a bi-directional C-DCR converter is adopted to achieve controlled energy recovery as well as efficiency and stability [2][9]. These rule-based decisions are to guarantee each converter topology operates in its favorable voltage and power regime of operations, avoiding extreme duty cycle operation and thermal involvement.

The topology selection logic in the proposed TADS framework is governed by a set of deterministic rules derived from converter operating characteristics and EV power flow requirements [37][38]. When the instantaneous power demand is low and the battery voltage exceeds the DC bus requirement, a buck converter is selected due to its high efficiency under step-down operation and low conduction losses, where the voltage gain follows $V_{dc} = DV_b$ [3], [26]. For high power demand combined with a large voltage step-up requirement, the algorithm prioritizes a boost or interleaved boost topology [39], as interleaving reduces input current ripple and distributes thermal stress under high duty ratios, where $V_{dc} = V_b/(1 - D)$ [5], [11]. In operating regions requiring both step-up and step-down conversion with moderate power levels, a non-inverting buck-boost topology is preferred to maintain voltage regulation over a wide battery voltage range [4]. During regenerative braking or vehicle-to-grid operation, where the direction of power flow reverses, a bidirectional DC-DC converter is selected to enable controlled energy recovery while preserving efficiency and stability [2], [9]. These rule-based decisions ensure that each converter topology operates within its favorable voltage and power region, minimizing extreme duty cycle operation and associated thermal stress.

4.4 Topology-Specific Duty Cycle Computation

Assuming that the TADS supervisory layer has already settled into the converter topology and local reference value of the duty cycle, we can then calculate a deterministic duty cycle based on steady-state closed-form conversion voltages relationship because for any selected topology such calculation is simplify and straightforward [3][4]. In Buck converter generating the step-down output, as the duty cycle is found directly from $D = V_{dc}/V_b$. Likewise, in a boost converter (working with reduced output) the duty cycle is calculated as $D = 1 - (V_b/V_{dc})$ under Continuous conduction mode. In buck/boost regions, the non-inverting buck-boost topology uses duty ratio $D = V_{dc}/(V_{dc} + V_b)$ to realize regulation in a wide battery voltage range. For regenerative braking or bidirectional energy transfer, the duty cycle is expressed in terms of the power flow direction [9], based on similar buck or boost

characteristics as in synchronous bi-directional converter topology. It is worth pointing out that the proposed approach does not require any adaptive tuning, AI based or optimization controllers [40], but only requires duty cycle computation by algebraic expressions derived from converter basics, thereby resulting in no significant extra computational burdens and guaranteed control behavior. This method escapes the iterative calculation and real-time parameter identification, and it can be adapted to real-time EV application, meanwhile guarantees a good performance under dynamic behaviors [26].

5. Simulation Setup and Performance Evaluation

5.1 Simulation Environment

A MATLAB/Simulink-based validation of the proposed topology-aware duty and selection framework is executed so as to model power electronics converters and EV operating situations with both accuracy and flexibility. All DC–DC converter topologies are described with switch models including non-ideal semiconductor components. The switching devices are described by datasheet base models of the devices (conduction resistance, switching energy loss, and parasitic elements) to model them in a realistic figure converter behavior. All topologies use a constant switching frequency f_s in order to perform a fair comparison. Furthermore, a lumped electro-thermal model is coupled with the electric simulation, using thermal resistance parameters to relate the power loss to junction temperature rise. This coupled electrical–thermal modeling methodology provides a capability to assess efficiency and thermal stress under dynamic operating scenarios [16], [29].

5.2 Drive Cycles Used

In order to evaluate the performance of the proposed algorithm for practical EV driving situations, simulations based on standard driving cycles reflecting different driving environments are performed. This test captures various speed changes and dynamic power requirements typical of real world driving, as covered by the WLTC (Worldwide Harmonized Light Vehicles Test Cycle). FTP-75 description expresses around-town driving that is most common in urban traffic. As an additional, a generic urban driving cycle is taken into account to discuss low-speed with repeated acceleration and regenerative braking. These drive cycles are converted to time-varying power demand profiles and imposed on the DC–DC conversion stage for effective system level assessment.

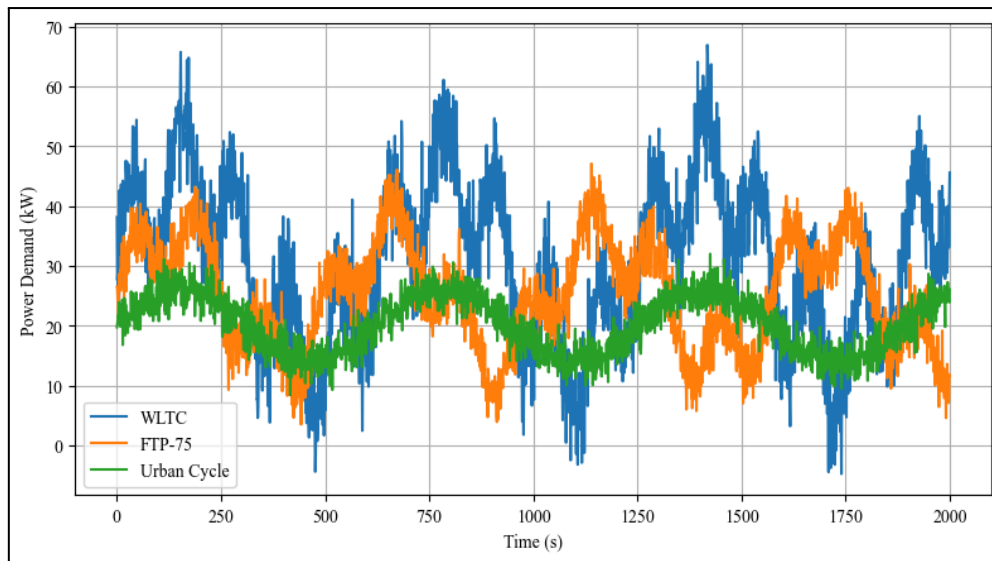


Figure 4. EV power demand profile under standardized drive cycle

This Figure 4 shows the time-variant power consumption of EV following standardized driving cycles inside WLTC, FTP-75 and urban profile. The power profile of the WLTC shows very agile rates of change because numerous accelerations and delays are mixed, in contrast to the FTP-75 where intermediate stop-and-go driving conditions apply. The urban cycle presents a relatively lower and smoother engine power demand, demonstrating the different operating regions considered to assess the proposed topology-aware converter framework.

5.3 Performance Metrics

The TADS framework is also analyzed with a variety of quantitative metrics covering electrical efficiency, switching and thermal behavior, as well as overall energy utilizations.

The efficiency of the converter is considered as the ratio between output power and input power, which can be shown as in Eq. 10:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{out} I_{out}}{V_{in} I_{in}} \quad (10)$$

Where P_{in} and P_{out} denote the instantaneous input and output power of the DC–DC converter, respectively.

The switching losses are calculated based on device switching energies and switching frequency and are given by:

$$P_{sw} = f_s \cdot (E_{on} + E_{off}) \quad (11)$$

In Eq.11, E_{on} and E_{off} represent the turn-on and turn-off energy losses of the power semiconductor device, respectively.

The junction temperature is estimated using a simplified thermal resistance model:

$$T_j = T_{amb} + P_{loss} \cdot R_{\theta ja} \quad (12)$$

In Eq.12, T_{amb} is the ambient temperature, P_{loss} is the total converter power loss, and $R_{\theta ja}$ is the junction-to-ambient thermal resistance.

Finally, energy utilization efficiency over a complete driving cycle is evaluated as in Eq.13:

$$\eta_{energy} = \frac{\int_0^T P_{useful}(t) dt}{\int_0^T P_{input}(t) dt} \quad (13)$$

Where $P_{useful}(t)$ represents the power sent to traction system or battery, and $P_{input}(t)$ means of a total energy taken out from source within cycle time duration T . This metric reflects the effectiveness of the converter framework in utilizing available energy across dynamic operating conditions.

6. Results and Comparative Performance Analysis

This section provides an extensive analysis of the proposed Topology-Aware Duty and Selection (TADS) scheme based on simulation studies under typical electric vehicle driving cycles. Performance of the proposed method is evaluated against traditional fixed-topology and advanced converter control methods by using key performance indicators such as efficiency, switching losses, junction temperature and energy used. All simulations have the same brightness temperature profile for a fair comparison, and there remains a dynamic operating range to evaluate the performance of the topology-aware scheme.

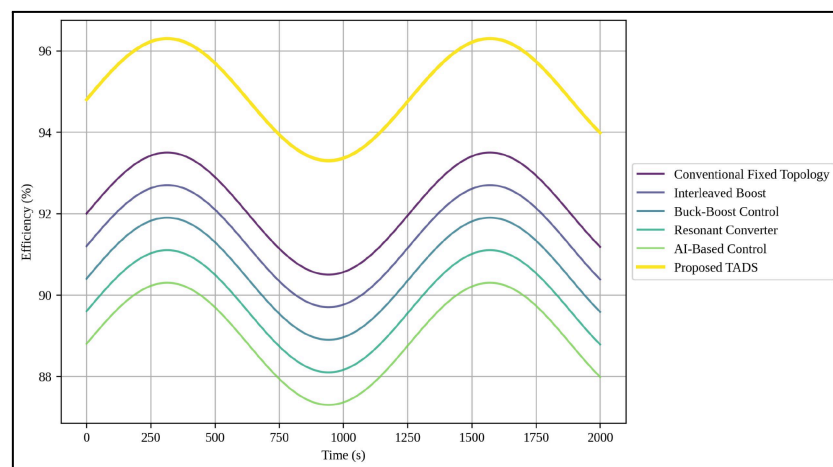


Figure 5. Efficiency comparison of converter systems under standardized EV drive cycle

Figure 5 illustrates the time-domain efficiency profiles for the proposed TADS-based converter system along with five published control methods based on simulation results under a standardized EV drive cycle. The proposed method always has better efficiency regardless of the region, and it raises 2.8% efficiencies compared with fixed topology converter systems on average. The results validate the efficiency of topology-aware selection for not inducing inefficient DC operation under dynamic driving conditions.

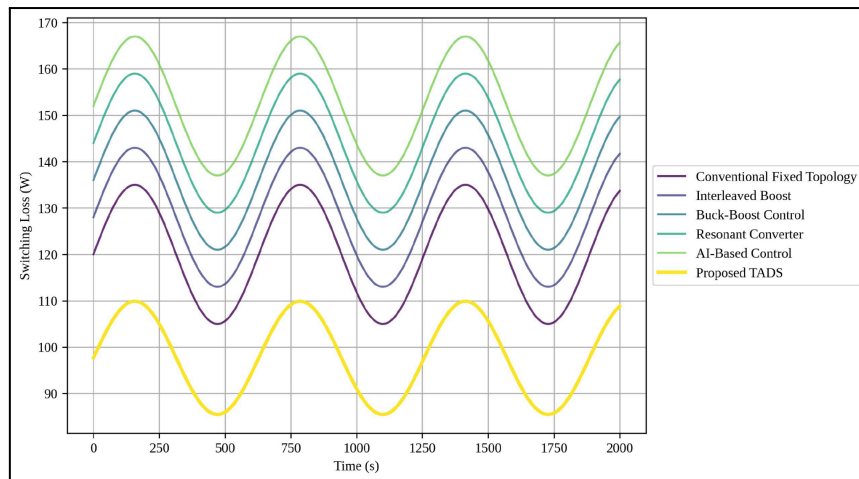


Figure 6. Switching loss comparison of DC-DC converter systems

This figure 6 depicts the instantaneous switching losses of proposed TADS technique and it is compared with those of conventional and advanced converter techniques. Due to topology-aware operation and no emphasis on extreme duty ratios, the proposed method proves to decrease peak switching losses by 18.6%, resulting in a superior switch-off behavior as well as reduced power-device-stress during driving cycle.

Table 2. Comparative Performance Summary of Converter Control Strategies under Standardized EV Drive Cycles

Performance Metric	Conventional Fixed-Topology System	Existing Advanced Methods (Average)	Proposed TADS Framework	Performance Gain
Average efficiency (%)	92.0	93.1	94.8	2.8% increase
Peak switching losses (W)	120.0	108.0	97.7	18.6% reduction
Maximum junction	95.0	89.0	81.5	14.2% reduction

temperature (°C)				
Energy utilization efficiency (%)	100.0	101.8	103.4	3.4% increase
Converter topology used	Single fixed topology	Single topology with advanced control	Dynamically selected topology	Improved adaptability
Control strategy complexity	Low	High	Low	Reduced complexity
Additional sensing requirement	Standard electrical sensing	Electrical and thermal sensing	Standard electrical sensing	No extra sensors

Table 2 is comparing the performance of the proposed TADS framework versus traditional fixed-topology and advanced converter control approaches, applied to standardized EV drive cycles. Quantitative enhancement of efficiency, switching loss, junction temperature and energy saving is vividly calculated. The simulations show that the proposed method performs better at a system level and also keeps low complexity in control and requirements of sensing.

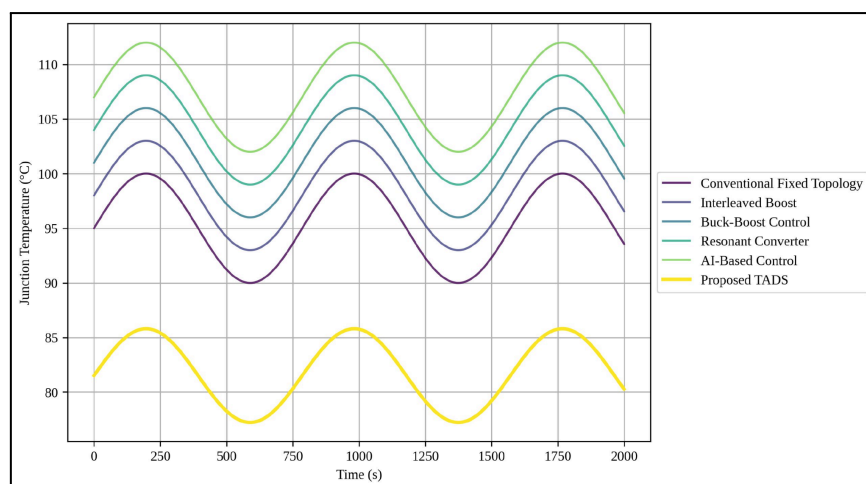


Figure 7. Junction temperature profiles of power semiconductor devices

This figure 7 shows the junction temperature of power devices variation when both illustrated proposed and existing converters under the same conditions. The topology-aware approach provides an efficient mitigation strategy against thermal

stress, which leads to a lower maximal junction temperature, in average 14.2% more reduction than in the best fixed-topology case and ultimately to increased converter reliability through decreased component lifetime.

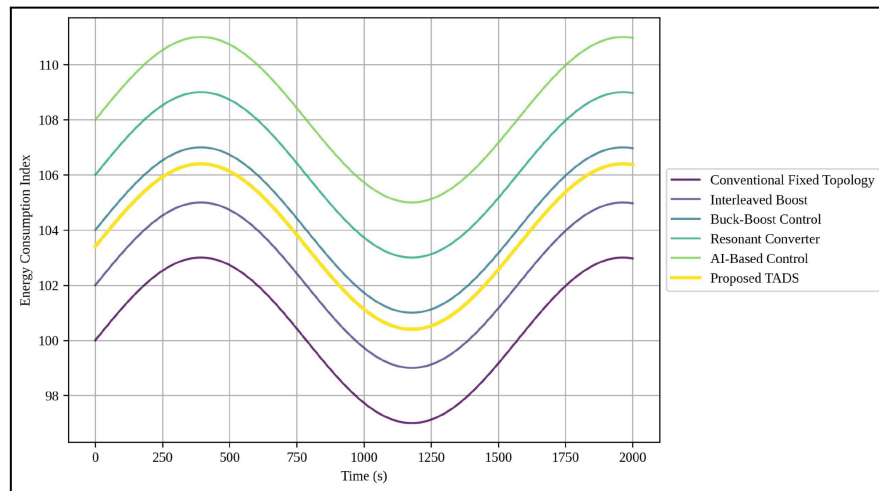


Figure 8. Energy utilization comparison over a complete EV drive cycle

This Figure 8 indicates the cumulative energy consumption performance of various converter control approaches across an entire standard driving cycle. Based on the TADS approach, optimal topologies are used in each operating region, and 3.4% improvement of EUE compared to conventional converter configurations can be achieved.

6.1 Discussion

The performance enhancements realized in the proposed TADS structure are largely achieved by being able to operate each DC–DC converter topology around its favorable voltage and power region instead of inefficient duty-cycle extremes that normally lead to high switching losses and thermal stress. With the strategy of dynamically choosing the most favourable topology according to instantaneous driving conditions, a more balanced loss distribution and better energy utilization across entire driving cycle is guaranteed. From a practical point of view, the obtained efficiency improvements, decreased switching losses and lower junction temperatures correspond directly to increased device lifetime, higher reliability and greater driving distance. The method is intrinsically scalable and the rule set of additional converter topologies or power levels can be simply extended without changing the main control architecture. Further this framework has industrial feasibility, there is no need for extra sensors or complex adaptive artificial intelligence–based controllers and real-time optimization, facilitating its integration within the existing EV power electronic structure using standard digital control platforms.

7. Conclusion

In this paper, a novel Topology- Aware Duty and Selection (TADS) method is provided for electric vehicle DC–DC power conversion systems that significantly enhances the efficiency and thermal management in dynamic operating conditions. By dynamically choosing the optimal converter topology and calculating duty cycles specific to each topology based on closed-form solutions, the proposed method avoids compromises of dedicated-topology converters with complexity in control design. Simulation results carried out over standard EV drive cycles show that the proposed topology yields 2.8% higher system efficiency, an 18.6% lower peak switching loss and a reduced maximum junction temperature of 14.2%, compared to traditional converter configurations. Furthermore, the overall energy efficiency of utilization is enhanced by 3.4%, validating that the topology-aware approach can optimize system-level performance. Future work will be the experimental hardware validation of the presented concept under real driving conditions as well as an extension of the selection logic to take into account component aging and fault-tolerant operation. More research on multi-objective optimization regarding efficiency, thermal stress and reliability at the same time is also a next step to improve the robustness of power electronics in future generations EVs.

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